

General Advice

Dr Cinar:

- Carrier performance not shown quantitatively. Most teams state that catapult/arresting constraints were applied, but I don't see min end speed checks at $WOD = 0$ (C-13-2) nor an arresting check with the Mk 7 Mod 3 curve at $WOD \leq 15$ kt. Put the point on the RFP curves at arrestment weight and report margins; include SEROC engine-out climb for multi-engine (≥ 200 ft/min at launch, ≥ 500 ft/min on approach).
- For a twin engine jet, the RFP requires SEROC ≥ 200 ft/min at launch and ≥ 500 ft/min on approach. Your constraint plot labels "SEROC approach"/"SEROC takeoff," but you don't show the actual single-engine climb-gradient computations at the launch end speed and at approach configuration/weight. Provide explicit SEROC calculations tied to the C-13-2 end-speed and Mark 7 approach speed per RFP Sec. 3.2.2.
- A modern FBW fighter can be statically unstable, but you still must manage CG travel and demonstrate controllability and trim across all required loading and Mach states (subsonic and supersonic). A "slightly unstable" SM will not pass "demonstrate stability for all flight and loading conditions" requirement in the RFP without a credible FBW/control-law concept and authority margins—especially at approach. If you keep the negative SM in the final design, show trim/control power and pitch-rate limits at catapult end speed as well.

Miranda:

Overall, the designs are developing well. The Q&A sessions identified some key considerations for teams to take into the coming weeks, and I look forward to seeing how that advice gets incorporated in your next assignments! Before the CDR, I would like to see more attention paid to carrier suitability – particularly to special considerations such as launch and arrest. The RFP includes some important performance requirements related to launch and arrest. Most teams did a good job of specifying these conditions on their sizing plots, but you also need to show that your design meets each requirement explicitly for the CDR. Also, when sizing for landing gear, don't forget that your aircraft will require a hood for carrier operations!

Common mistakes for this assignment included neglecting the internal layout drawing, missing performance requirements and sizing plot conditions (see slides on RFP requirements), and failing to include the center of gravity and neutral point on the 3-view drawing. Another common loss of points related to insufficient discussions of stability, particularly missing tail volume coefficients.

As you move your designs forward, consider where it makes sense to continue using historical values. The designs in class are deviating further and further from the historical trends you collected at the start of the semester (which is a good thing!), so you may need to check where the assumptions that underly that data are still valid. For example, if your airfoil or wing designs are different from your comparison aircraft, then guiding your tail sizing by those aircraft may not be the best approach. Don't be afraid to consider a wider design space as you explore more detailed analyses in the coming weeks.

Dillon (former 481 student):

All of the teams did a good job and it's always impressive to see what others come up with for these projects, especially having worked on a similar one myself recently. I was a bit surprised to not see more about the aircraft-carrier catapult takeoff/landing or material selection; these were the things that immediately stood out to me as challenges distinct to this year's RFP. I also wish they talked more about what makes their aircraft different from the historical aircraft in their comparison tables, referencing some stand-out numbers that we should be aware of. I have some specific observations about this below. Final thing, I was hoping for some more justification of exactly why they chose their respective design points on the constraint diagrams. Obviously these are big feasible regions so I'm wondering what made them pick their specific point.

I also think it would be helpful (as I mentioned in class) to show operating points (cruise, landing, combat, dash, etc.) when giving something like a drag polar: it's great they can estimate their drag for every CL but they're not actually flying at every CL.

Team 1

Dr Cinar:

- Slide 7 - it's not clear which constraints apply to low-thrust vs military thrust. Consider braking it down to two plots in the future. Otherwise, great job covering these
- Slide 9 - T vs S slide shows total thrust = 29,784 lbf. Is this for one engine? Your design point has a T of about double this amount, so I'm assuming yes. Double check. T in a T-S diagram must be installed net total at the relevant flight condition.
- Slide 12 - great summary. Missing your logo!
 - "2 engines selected" -- this should be a factor of cost too. It'd be nice to have a comparison of all engines you considered and why you chose to go with the one you did
- Slide 17 -- great job showing this trade!
- Slide 18 -- Wing planform and airfoil optimization are well documented; polars for multiple flap/gear states are included. Once you're done with your supersonic flow analysis, you'll end up with an airfoil close to a diamond-shape. Looks like you've already considered the supersonic flow effects, great!
- Slide 23 - watch out for SM, and how it changes in different flight conditions. Your table shows -12 to -15% MAC static margin for the empty and payload+rear-fuselage fuel states, with only 0-5% for some fueled conditions (subsonic), and you add an unexplained "+25.2% for supersonic SM. That magnitude of NP shift and the empty tail-heavy condition are red flags. A modern FBW fighter can be statically unstable, but you still must manage CG travel and demonstrate controllability and trim across all required loading and Mach states (subsonic and supersonic). Re-work CG management (fuel and stores near the CG), wing/HT placement, and/or tail volume/arm; align empty, payload, and fuel CGs. Document subsonic and supersonic NP derivations
- Great use of your CAD drawings throughout the presentation
- Presentation is very well executed, logical flow & smooth transitions. Everybody presented well
- *You should seriously consider sending your design to the AIAA competition*
- ~18 min presentation time
- Handled the Q&A very well.
- Other notes:
- Carrier SEROC not demonstrated. For a twin, the RFP requires SEROC ≥ 200 ft/min at launch and ≥ 500 ft/min on approach. Your constraint plot labels SEROC

approach / SEROC takeoff, but you don't show the actual single-engine climb-gradient computations at the launch end speed and at approach configuration/weight. Provide explicit SEROC calculations tied to the C-13-2 end-speed and Mark 7 approach speed per RFP Sec. 3.2.2.

- Catapult and recovery checks are not traceable to the RFP curves.
- The RFP provides C-13-2 and Mark 7 Mod 3 performance charts; the CDR must show (a) end-of-stroke speed requirement vs MTOW and CLmax (90% rule) and (b) recovery (VAPP/arrestment energy) with WOD limits. Put your numbers directly on those charts, not just as generic catapult/recovery constraints
- Recovery must be done at landing weight per the RFP's rule (fuel for 20-min loiter + two attempts, $\leq 25\%$ fuel max, 50% stores) and $VAPP < 145$ kt with $VAPP \geq 1.1 V_{stall}$. Re-compute and show weight, CLmax(LDG), V_{stall} , VAPP, and arrestment speed (I see your landing weight assumption in the backup slide 39, but takeoff and landing is a significant part of the design drivers so we want to see the performance under the given requirements in the RFP.
- Include important metrics like Cruise L/D, Static margin for your aircraft, and cruise SFC for comparables if available, in your big comparison table
- For the Dash 2 drawings: Your component locations list percentages for bay and tanks but you do not show where $6 \times$ AIM-120 + $2 \times$ AIM-9X and $4 \times$ Mk-83 actually fit. Add internal bay drawings with dimensions and an external-stores arrangement for both missions on the 3-view page, and account for their CG effects through the mission
- RFP asks for comparable unit acquisition cost with a credible parametric basis. Show your cost parametrics, learning curve, content assumptions (materials, avionics, engine, profit), and index everything to 2025 USD.

Miranda:

- Good explanation that you are maximizing the dimensions, and justification of folded configuration dimensions!
- Great explanation of your flight implications and avionics
- Take a closer look at your aero analysis, having the cruise (ie most efficient) condition be at the location of lift discontinuity could provide significant instabilities in flight.
- Mission profiles: take a look at your descent stages, and make sure you are breaking the climb into appropriate segments.
- Sizing Plot - Ceiling and climb constraints are generally horizontal lines, so it may be worth double checking your code. That said, your design point appears reasonable.

- CAD Drawings
 - Missing internal 3-view drawing, although good inclusion of the CAD for the pilot sight lines and detailed component drawings.
 - Need to include center of gravity and neutral point on the drawings
- Missing discussion of SEROC conditions, crew weight, and clearly defined cruise conditions (Mach #, altitude) of the design cruise condition